

FFN for Language Model:N gram model comparison (Song generator)

Built an FNN-based language model in PyTorch from scratch: tokenization, embedding layers, n-gram context pairs, CrossEntropyLoss training loop, t-SNE , and text generation with temperature sampling. Interesting part was why it failed :

Finding 1: The 8-gram model hit perplexity ~ 1.0 largely due to small corpus

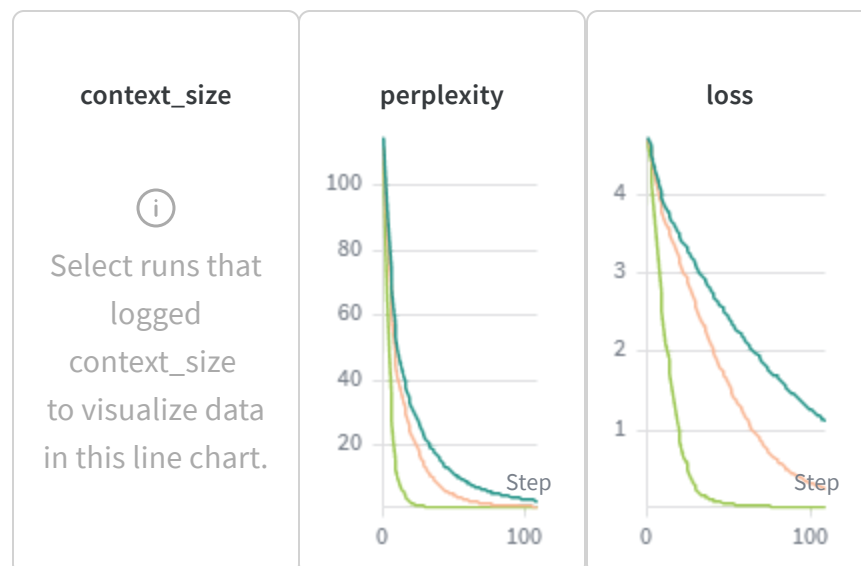
Finding 2: The coherence wall. All three context sizes (N=2, 4, 8) collapsed into repetitive loops within 5-8 words

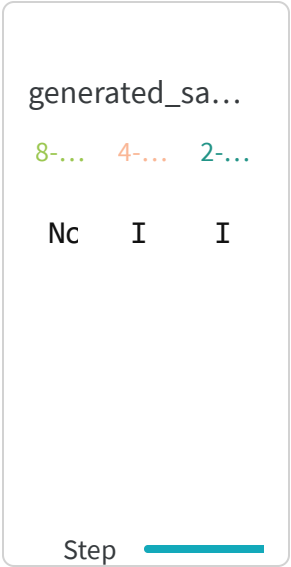
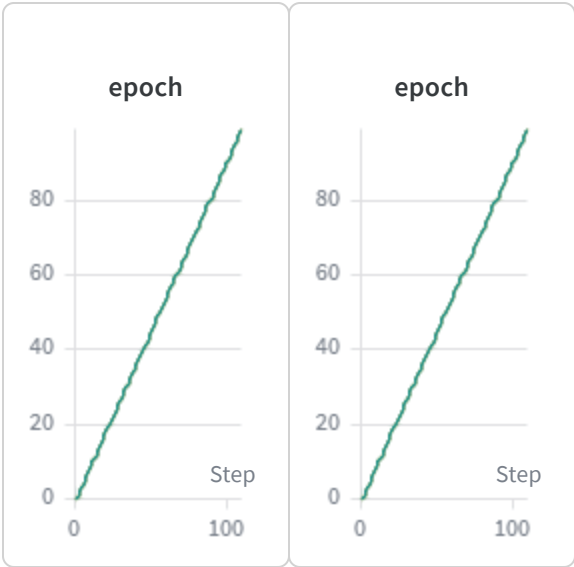
Conclusion: Even with 1 head transformer with attention could deliver better

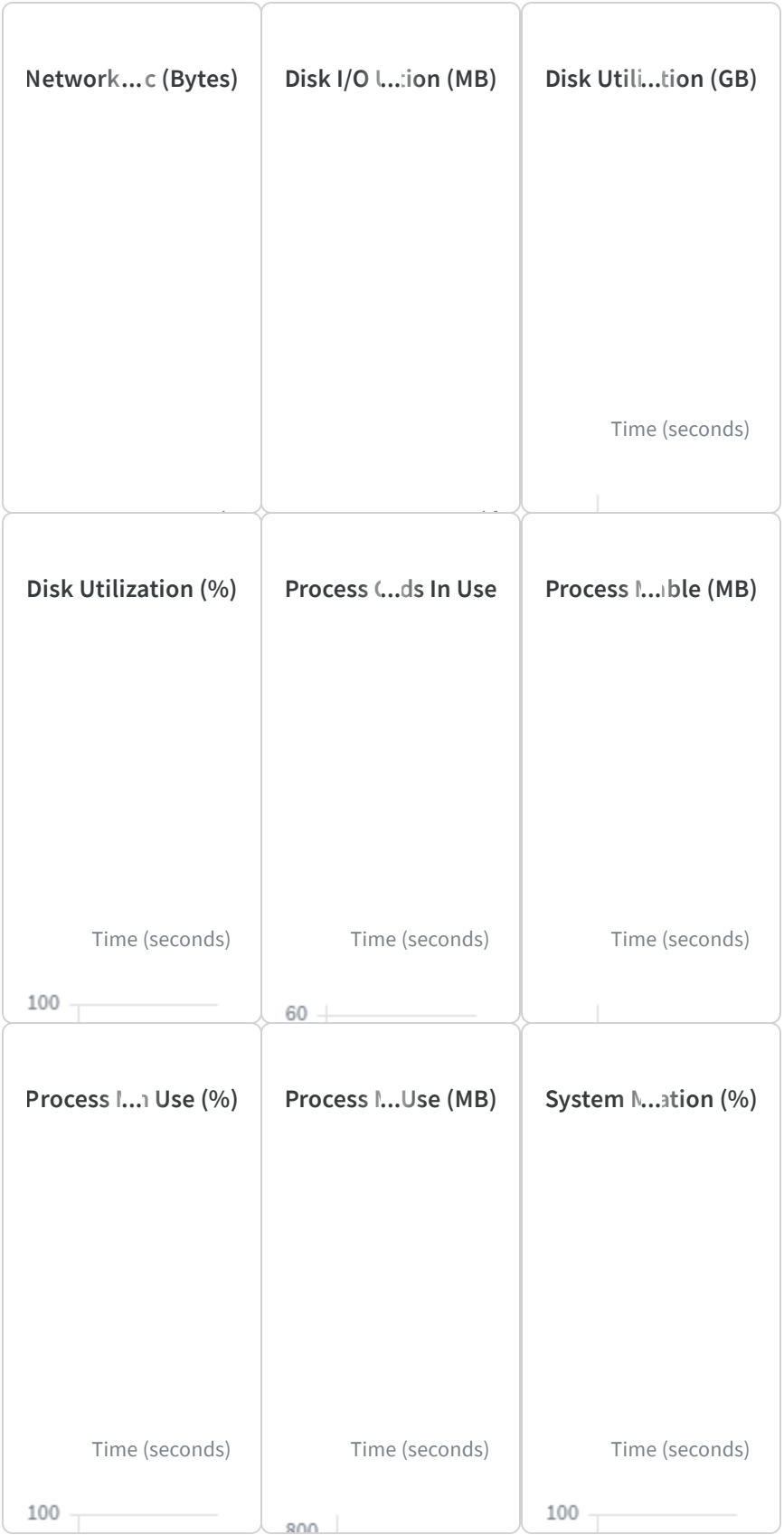
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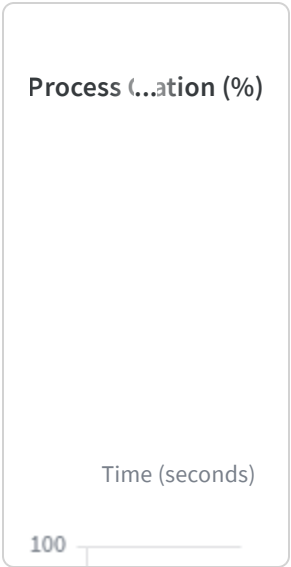
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✓ Section 1









Created with ❤️ on Weights & Biases.

<https://wandb.ai/nishanth-anantha87-asml/ngram-song-generator/reports/FFN-for-Language-Model-N-gram-model-comparison-Song-generator---VmldzoxNjUwMzA4NA>

